↳ Any one of 20 students can get 1st rank in one subject. So, there are 20 ways to get 1st rank by 20 students. In the same way in the other 2 subjects too.

$$\underset{M}{20} \quad \underset{P}{20} \quad \underset{C}{20} = 20 \times 20 \times 20 = 20^3 = n^r.$$

$n = 20 =$ no. of objects, $3 =$ no. of places $= r$.

Permutations (with repetition):- 'n' objects are to be placed in 'r' places ($r \leq n$) with repetition in $\boxed{n^r \text{ ways}}$.

Rules of Permutation:-

(i) ${}^nP_1$:- ${}^nP_1 = \frac{n!}{(n-1)!} = n$ ways

Ex:- How many ways we can give royal chair to 5 members.

Ans:- ${}^5P_1 = 5$ ways.

(ii) ${}^nP_0 = \frac{n!}{(n-0)!} = 1$

(iii) ${}^nP_n = \frac{n!}{(n-n)!} = \frac{n!}{0!} = n!$

Ex:- How many ways we can arrange 3 students in 3 ranks of a subject?

$$\underset{1st}{3} \quad \underset{2nd}{2} \quad \underset{3rd}{1} = 6 = {}^3P_3 = 3! = \underline{6}.$$

(iv) ${}^nP_{n-1} = \frac{n!}{(n-(n-1))!} = \frac{n!}{(n-n+1)!} = n!$

Ex:- How many ways we can arrange a, b & c in 2 places?

$${}^nP_r = {}^3P_2 = \frac{3!}{(3-2)!} = 3! = 6 \text{ ways}$$

ab, ac, ba, bc, ca, & cb

here I already told each arrangement is different- by considering order of objects in the arrangements

$ab \neq ba,\ ac \neq ca,\ bc \neq cb$

Notes:- (i) 'n' ~~lettered~~ no distinct lettered word can be arranged in 'n!' ways.

(ii) In a 'n' lettered word if one letter is 'a' times, another letter is for 'b' times ..... then no of arra- -ngements is $\frac{n!}{a! \cdot b! \cdot c! \cdots}$

(iii) 'n' distinct letter word can be arranged in such a way that 'k' letters are together in.

$(n-k+1)!\, k!$ ways.

(iv) 'n' distinct lettered word can be arranged in such a way that 'k' letters are not together in

Total ways - together ways = not together ways.

$[n! - (n-k+1)!\, k!]$ ways.

↳ Circular Permutation:- If the objects are to be arran- -ged in circular fashion then Permutations are $(n-1)!$ ways.

↳ Combinations:- Selection of objects from given objects in a particular no. of ways based on condition in the question are combinations.

Combination:- Selection of 'r' objects ($r \le n$) from 'n' objects in nC_r ways.

$$\therefore nC_r = C(n,r) = \frac{n!}{(n-r)! \cdot r!}$$

↳ Important properties:-

(i) $nC_0 = \frac{n!}{(n-0)! \cdot 0!} = 1$ [The no. of ways to select '0' objects from 'n' distinct objects]

(ii) The no. of ways to select exactly one object out of 'n' distinct objects is,

$$nC_1 = \frac{n!}{(n-1)! \cdot 1!} = \frac{n(n-1)!}{(n-1)!} = n$$

(iii) The no. of ways to select 'all' the objects out of 'n' distinct objects is nC_n

$$nC_n = \frac{n!}{(n-n)! \cdot n!} = 1$$

(iv) The number of ways to select 'r' objects out of n objects is equals to the no. of selections of 'n-r' objects out of 'n' objects.

i.e., $nC_r = nC_{(n-r)} = \frac{n!}{(n-r)! \cdot r!} = \frac{n!}{(n-(n-r)) \cdot (n-r)!}$

$= nC_r$.